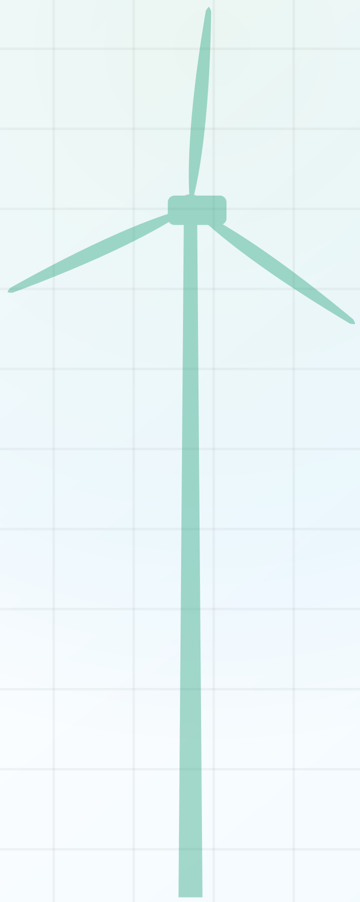
SCOTLAND · WIND ENERGY · COMPUTE · CLEANTECH

Scotland's Wasted Wind. The UK's Greenest Compute.

WindEdge co-locates modular GPU data centres at wind farm substations — converting curtailed energy into the cheapest, most verifiably renewable compute in the UK.

Thomas Travis · CEO | Sulayman Awan · CTO | Linus Deluca-Perry · Strategy | Ye He · Engineering



THE PROBLEM

Two Broken Markets. One Structural Fix.

Scotland generates wind energy and throws it away. UK enterprises need compute they can't afford to power cleanly. These are the same problem.

⚡ THE CURTAILMENT CRISIS

9 TWh
wind energy wasted in Scotland annually (2025)

£350m
paid to wind farms to switch off (2025)

↑ YoY
growing as wind capacity outpaces grid upgrades

Scotland has ~17.7 GW of installed wind capacity. The Anglo-Scottish interconnector cannot carry the power south — so National Grid pays wind farms to **switch off**, then pays gas plants in England to generate instead. This is structural, worsening, and profitable to solve.

🖥 THE DATA CENTRE ENERGY PROBLEM

£140/MWh
electricity cost for UK data centres

40–60%
of total opex is electricity

2026
UK SDR mandates Scope 2 emissions reporting

From 2026, UK Sustainability Disclosure Requirements force enterprises to report Scope 2 emissions. Renewable Energy Certificates are **widely criticised as greenwashing**. There is no UK product that offers genuinely renewable compute — power proven to come from a turbine, not a certificate.

THE INSIGHT

The wasted wind IS the product.

Wind farms in high-curtailment areas have a substation, land, and power available at near-zero cost — because that power would otherwise be discarded.

ENERGY COST COMPARISON

OPERATOR TYPE	COST / MWH
Typical UK colocation	£100–280
WindEdge (curtailed wind)	**PPA cost unknown — but energy prices in high-curtailment regions regularly go negative**

Structurally lower

energy cost vs grid — curtailment prices can go negative where WindEdge operates

THE GREEN PREMIUM

Renewable Energy Certificates

Certificates claim renewable generation *somewhere*, *sometime*. Widely criticised. Cannot satisfy 24/7 matching requirements. Equinix, Digital Realty, iomart all rely on these.

WindEdge: At-Source Direct Wire

The electrons running through the server came from the turbine 200 metres away. **This is the strongest possible green claim** — it cannot be replicated by any grid-connected operator at any price. Enterprise buyers pay a **10–25% premium** for verifiable at-source renewable compute.

THE SOLUTION

Modular GPU Data Centres at Wind Farm Substations

WindEdge deploys containerised facilities directly at the substation — consuming power that would otherwise be curtailed, pairing with battery storage for continuous uptime.

Wind Turbines
17.7 GW installed (Scotland)

→

MV Substation
Direct-wire connection

→

Battery Storage
Up to 2 MWh per site (20 hrs storage)

→

WindEdge Data Centre
100 kW · ~60 GPUs per module

→

Enterprise Customers
AI · HPC · Colocation

Why Containers

12–16 week delivery (vs 18–36 months for traditional builds). Permitted development rights on wind farm land. Each container is a discrete, mortgageable asset. Scotland's ~8°C climate enables free-air cooling for ~89% of operating hours — targeting PUE 1.05–1.10 vs. industry average of 1.58.

Grid Tie-In for Backup

4-layer power resilience: curtailed wind → battery → grid → diesel genset. Colocation SLAs guaranteed at >99.9% uptime despite wind intermittency.

Customer Value: Enterprise Colo

10–25% below-market pricing · PUE 1.05–1.10 · Real-time carbon dashboard · Zero Scope 2 emissions — not offsets

Customer Value: HPC / AI Compute

Managed GPU clusters at 40–50% below hyperscaler rates · Green AI credentials · UK data sovereignty · GDPR compliant

Wind Farm Partner Value

Revenue from energy that would be discarded · Revenue share or equity stake · Local employment

CORE IP

The Energy Management System

A real-time multi-objective optimisation algorithm — WindEdge's primary patent candidate. Makes intermittent wind compatible with commercial SLAs.

Inputs → Outputs

Inputs: Wind generation forecast · Turbine SCADA data · Battery state of charge · GPU marketplace demand · Grid spot price

Outputs: Battery charge/discharge commands · GPU load scheduling · Grid import/export decisions

LOAD PRIORITY HIERARCHY

PRIORITY 1	**Colocation Customers**	Always on — SLA-protected
PRIORITY 2	**Managed HPC Jobs**	Paused / rescheduled
PRIORITY 3	**GPU Training Queues**	Delay-tolerant
PRIORITY 4	**Flexible Mining**	Instant-off — absorbs surplus

Up to 2 MWh battery (20 hrs storage) bridges wind lull events. Grid covers extended low-wind periods. Result: >99.9% uptime for colo customers.

WHY NOW

Five Converging Forces

Scottish curtailment growing year-on-year	More sites available, lower energy cost — the opportunity improves with time
GPU compute demand at record highs	Higher marketplace rates, stronger utilisation, longer contract terms
Mandatory Scope 2 reporting from 2026	UK SDR creates real willingness to pay for verifiable green compute
RTX 5090 affordable enough to bootstrap	Phase 0 pilot provable for ~£12,500 — real data before investors
Zero direct UK competitors	No UK operator co-locates modular data centres with wind farms today

BUSINESS MODEL

Three Phases. Capital-Efficient by Design.

Start with a self-funded pilot to prove revenue. Build managed facilities to generate cash flow. Own the best sites to capture full margin.

PHASE 0 · NOW

Self-Funded Pilot

Buy **4 × RTX 5090 GPUs** (~£10,000) plus server components (~£2,500) — ~£12,500 total. List on Vast.ai and RunPod. Generate real GPU marketplace revenue — no wind farm, no investors required.

Proves the model with real data before any external capital is raised. Estimated **~$0.40/GPU/hr net** at 85% occupancy. Direct customer escalation from anonymous marketplace listings to contracted relationships with startups, universities, and AI teams.

~£9,400/yr

Estimated pilot net revenue (4 GPUs · 85% occupancy) · ~1.5yr payback

PHASE 1 · PRIMARY MODEL

Build-and-Manage Service

WindEdge designs and commissions **100 kW modular GPU facilities** for clients — wind farm operators, landowners, private investors. Client owns the hardware; WindEdge provides the expertise and ongoing management.

Revenue: Fixed build fee (10% of total build cost) + 10% of gross GPU revenue monthly. Clients get a passive income stream from an asset they own. WindEdge gets recurring revenue without owning depreciating hardware.

£150k/site/yr

WindEdge 10% share of £1.5m GPU revenue per 100 kW managed facility

PHASE 2 · SCALE

Owned Facilities

At the highest-curtailment Scottish sites, WindEdge owns and operates facilities directly — capturing **100% of GPU revenue** rather than 10%. Two routes: funded from accumulated build-and-manage revenue, or accelerated via a **seed raise**.

A 20-site Scottish portfolio (20 × 100 kW managed) represents **£30m+ addressable revenue**. Owned facilities generate the majority of profit at steady state.

£30m+

Addressable revenue from 20-site Scottish portfolio

UNIT ECONOMICS

Strong Numbers at Every Scale

Economics driven by structural energy cost advantage — not subsidies or market cycles.

PER 100 KW MANAGED FACILITY

£1.5m
Annual GPU revenue potential

£150k
WindEdge 10% management fee

+10%
One-off build fee (10% of total build cost) on completion

SCALE

£15m/yr
GPU revenue from a 1 MW facility

£30m+
Addressable revenue, 20-site Scottish portfolio

ENERGY COST ADVANTAGE (OWNED FACILITY, PER YEAR)

COST ITEM	WINDEDGE	GRID-CONNECTED COMPETITOR
Energy	PPA cost unknown — prices regularly go negative in curtailment regions	£100–280/MWh
Cooling (PUE)	1.05–1.10	1.40–1.60
Transmission charges	£0 (behind-meter)	£15–25/MWh
Total effective cost	**~£15–35/MWh**	**£100–280/MWh**

PHASE 0 PILOT

4 × RTX 5090 GPUs + server components · ~£12,500 total · Vast.ai / RunPod · ~$0.40/GPU/hr net · 85% occupancy · **~£9,400/year** · ~1.5yr payback. Self-funded — real revenue data before any investor capital is deployed.

MARKET OPPORTUNITY

A Growing Market. Zero Direct Competitors.

WindEdge sits at the intersection of two growing markets. No UK operator is executing this model today.

£8bn
UK data centre market today
→ £17bn by 2030

11%
CAGR global data centre market

19%
CAGR green data centre market (vs 11% conventional — nearly double the rate)

Target Segments

AI / HPC startups & researchers — GPU scarcity, cost sensitivity, green AI narrative

Enterprise sustainability leads — Scope 2 reporting obligation from 2026 (UK SDR)

Scottish public sector — NHS, universities, Scottish Government (data sovereignty + green mandate)

Private investors / family offices — 100 kW facility at ~£200–350k all-in; tangible AI infrastructure asset generating monthly GPU income

COMPETITIVE LANDSCAPE

COMPETITOR	MODEL	WEAKNESS VS. WINDEDGE
Equinix / Digital Realty	Large-scale urban colo	Grid energy, high PUE, no green premium
iomart	Scottish cloud provider	Legacy infrastructure, certificates only
Deep Green (UK)	Waste heat DCs	Cannot match wind curtailment scale
Hyperscalers (AWS, Azure)	Everything	Need 100 MW+ sites; ignore <5 MW gaps
UK wind-adjacent DC operator	**None exist. WindEdge is first.**	

International analogues validate the model: **AtNorth** (Nordic green DC, acquired by Equinix for $4.2bn), **GreenMountain** (Norway, acquired for $800m), **EcoDataCenter** (Sweden, €1.8bn raised), **Hydro66** (Sweden, acquired by Northern Data AG for €31.5m), **Borealis** (Iceland, geothermal). UK market is entirely unserved.

TRACTION & ROADMAP

Strong Foundations Before a Pound of Investment

Every significant pre-capital milestone has been completed. Phase 0 pilot begins immediately.

✓ COMPLETED

- ✓ Full technical architecture and electrical topology
- ✓ EMS algorithm concept and specification (patent candidate identified)
- ✓ Scottish planning and grid connection regulatory mapping
- ✓ Competitive landscape analysis — zero UK direct competitors confirmed
- ✓ Wind farm operator target list compiled
- ✓ IP strategy with prior art search commissioned
- ✓ Customer discovery with institutional HPC leads (Imperial College)
- ✓ Financial model with full 5-year P&L projection

→ NEXT STEPS

Phase 0 — Immediate

Purchase 4 × RTX 5090 GPUs (~£10,000) plus server components (~£2,500) — ~£12,500 total. List on Vast.ai and RunPod. Run 24/7. Document utilisation and revenue per GPU-hour for 3 months. Estimated ~£9,400/year at 85% occupancy. **Real revenue data before any external capital is deployed.**

Phase 1 — Months 1–18

First wind farm LOI · Grid connection feasibility study · First 100 kW managed facility built and operational · First direct customer contract signed

Phase 2 — Months 18–36

6 sites operational · Managed GPU product launched · Demand response registered with National Grid ESO · Series A fundraise (£18m target, Month 30 close) · Path to 20-site Scottish portfolio

THE TEAM

Built for Exactly This Problem

HPC infrastructure · Energy markets · Carbon trading · Frontier ML · Institutional finance · Renewable energy entrepreneurship. Three of us process petabytes of data daily — we are the customer.

TT

Thomas Travis

CEO & CO-FOUNDER

PhD in particle physics, CERN / Imperial College London. Leading the full Run 3 VBF Higgs-to-invisible analysis on CMS — a flagship dark matter search — using Lorentz-boosted neural networks on petabytes of LHC collision data.

Shell Oil Trading (2x internship) · Energy PPAs · CERN LHC Grid · Commercial Strategy

SA

Sulayman Awan

CTO & CO-FOUNDER

PhD in particle physics, CERN / Imperial College London. Pioneering ML research using Graph Transformer networks for model-agnostic dark matter searches. Quant research at Moore Capital; carbon trading strategy at Altana Wealth.

Moore Capital (quant) · Altana Wealth (carbon) · EMS Algorithm · GPU Infra

LD

Linus Deluca-Perry

FOUNDING BUSINESS STRATEGIST

Business Analyst at BlackRock's Transformations Office, leading large-scale change programmes. Physics BSc, Imperial College London. Founder of LDP Solar — independently developed and patented an agri-solar mounting system.

BlackRock · LDP Solar (founder) · Patent holder · Investor Relations

YH

Ye He

FOUNDING ENGINEER & TECHNICAL ADVISOR

PhD in particle physics at CERN / Imperial College London, measuring the top quark mass using simulation-based inference on CMS — the first application of this technique. Builds GPU-accelerated analysis workflows for large-scale datasets across the CMS collaboration.

CERN CMS · GPU Pipelines · HPC Workflows · ML Research

Thomas, Sulayman, and Ye process petabytes of collision data daily on the CERN Worldwide LHC Computing Grid — **they are the exact customer WindEdge serves**, and they understand first-hand what GPU compute costs and why green provenance matters. Linus has already taken a renewable energy technology from concept to patent to pilot.

THE ASK

£2.4m Seed Round · Target Close Q4 2026

Raised after the Phase 0 pilot has generated real revenue data — de-risking the investor conversation before it begins.

USE OF FUNDS

First owned 100 kW wind-adjacent facility	**~£300,000**
Battery storage (up to 2 MWh)	**£80,000–100,000**
Legal — PPA, ground lease, incorporation	**£60,000–80,000**
Grid connection & feasibility	**£40,000–60,000**
Working capital (18 months runway)	**£120,000–160,000**
Total Seed Round	**£2.4m**

Build fees (10% of total build cost) fund WindEdge operations from Year 1. Series A (£18m, Month 30) accelerates rollout to 12 sites.

WHY INVEST NOW

SEIS/EIS

50% (SEIS) or 30% (EIS) income tax relief + CGT exemption for UK investors

1st Mover

Wind farm partnerships are exclusive per site — first operator wins it permanently

Defensive Moat

Curtailment grows as Scotland installs more wind. Competitors cannot replicate WindEdge's energy cost without replicating years of site-by-site wind farm relationships. EMS algorithm is IP-protected.

Market Validation — Comparable Companies

AtNorth acq. by Equinix $4.2bn · **GreenMountain** acq. for $800m · **EcoDataCenter** €1.8bn raised · **Deep Green** £200m raised · **Etix Everywhere** €200m raised · **Hydro66** acq. by Northern Data €31.5m. Infrastructure fund acquisition at 12–18× EBITDA implies a path to **£24–36m enterprise value** within 5–7 years.

WINDEDGE DATA CENTRES

Scotland's Wasted Wind.
The World's Greenest Compute.

We are green compute infrastructure for enterprises that turns Scotland's curtailed wind into the UK's cheapest and most verifiably renewable data centre capacity.

Thomas Travis — CEO

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£2.4m Seed Round Open | SEIS / EIS Eligible | Q4 2026 Close | Zero UK Competitors